

A QUICK GLOBAL MEASURE OF PERFORMANCE OF KERNEL DENSITY ESTIMATORS

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Summary: Asymptotic normality is established for a quickly
computable global measure of performance of kernel
density estimators which requires the evaluation
of the density estimator only on a subset of the
sample.

Let $X_1, \dots, X_n$ be independent and identically distributed random variables with absolutely continuous distribution function and denote by f the pertaining density.

The popular kernel density estimator of f , introduced by Rosenblatt (1956) and Parzen (1962), is given by

$$f_n(x) := (n\alpha_n)^{-1} \sum_{i=1}^n k((x-X_i)/\alpha_n)$$

where $\int k(x)dx = 1$ and $\alpha_n, n \in \mathbb{N}$, is a sequence of positive constants tending to zero as n increases. For surveys we refer to the articles by Scott, *et al.* (1977) and Wertz (1978).

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Suppose throughout this article that f is positive on $[0,1]$, say. The particular global measure of performance of kernel density estimators

$$T_n := \int_0^1 (f_n(x)/f(x) - 1)^2 f(x) dx$$

which can be regarded as a χ^2 -statistic with infinitely many cells, was investigated by Bickel and Rosenblatt (1973) and by Falk (1985).

Under the assumption that f is twice differentiable on $\mathbb{R}$ with bounded second derivative, Bickel and Rosenblatt (1973) proved asymptotic normality of T_n for sequences α_n , $n \in \mathbb{N}$, ranging between $n^{-1/4}$ and $n^{-2/9}$, roughly.

Using a different approach by representing T_n as a U-statistic, Falk (1985) established asymptotic normality of T_n for α_n satisfying $n\alpha_n^5 \xrightarrow[n \in \mathbb{N}]{\rightarrow \infty}$ which includes certain optimal bandwidths.

However, one practical disadvantage of T_n is obvious: It is in general not easy to calculate for a particular sample and one usually has to apply numerical methods to compute its value approximately.

To overcome this problem, we suggest in the present paper the following easy computable approximation S_n of T_n which requires the evaluation of the density estimator only on a subset of the sample $X_1, \dots, X_n$.

Put for $i=1, \dots, n$

$$f_{n,i}(x) := [(n-1)\alpha_n]^{-1} \sum_{j=1; j \neq i}^n k((x-X_j)/\alpha_n)$$

and define

$$S_n := n^{-1} \sum_{i=1}^n \left\{ f_{n,i}(X_i)/f(X_i) - 1 \right\}^2 1_{[0,1]}(X_i)$$

where 1_A denotes the indicator function of a set A .

In lemma 2 we will prove that S_n and T_n are stochastically equivalent

and thus, S_n can be regarded as a quick global measure of performance of kernel density estimators.

The key to the results in Falk (1985) was the following representation of T_n as a U-statistic

$$\begin{aligned}
 (1) \quad T_n &= n^{-2} \sum_{i,j=1; i \neq j}^n \int_0^1 \left\{ \alpha_n^{-1} k((x-x_i)/\alpha_n)/f(x) - 1 \right\} \cdot \left\{ \alpha_n^{-1} k((x-x_j)/\alpha_n)/f(x) - 1 \right\} f(x) dx \\
 &\quad + n^{-2} \sum_{i=1}^n \int_0^1 \left\{ \alpha_n^{-1} k((x-x_i)/\alpha_n)/f(x) - 1 \right\}^2 f(x) dx \\
 &=: n^{-2} \sum_{i,j=1; i \neq j}^n U_n(X_i, X_j) + n^{-2} \sum_{i=1}^n g_n(X_i).
 \end{aligned}$$

In the proof of the following basic result we will make extensive use of the representation of T_n and an analogous one for S_n .

Lemma 2. Suppose that f is twice differentiable on $\mathbb{R}$ with bounded second derivative. Let k be bounded and satisfy $\int k(x) dx = 1$, $\int x k(x) dx = 0$, $\int x^2 |k(x)| dx < \infty$ and $\int k^2(x) dx < \infty$. Then, for any $\epsilon > 0$ if $\alpha_n \rightarrow 0$ and $n\alpha_n^3 \rightarrow \infty$ as $n \in \mathbb{N}$

$$P\{n^{1/2} \alpha_n^{-2} |T_n - S_n| \geq \epsilon\} \xrightarrow{n \in \mathbb{N}} 0.$$

Put

$$\mu_n := E(U_n(X_1, X_2)) \quad \text{and} \quad \nu_n := E(g_n(X_1)).$$

The asymptotic values of μ_n and ν_n are given in lemmas 2.5 and 2.6 in Falk (1985) and the following result is now immediate from the preceding lemma and theorem 2.4 in Falk (1985). By $\rightarrow_p$ we denote convergence in probability

distribution and by $N_{(0,1)}$ the standard normal distribution.

Theorem 3. If in addition to the assumptions of lemma 2 $n\alpha_n^5 \rightarrow \infty$, then

$$n^{1/2} \alpha_n^{-2} c^{-1} (S_n - \mu_n - n^{-1} \nu_n) \xrightarrow{D} N_{(0,1)}$$

$$\text{where } c^2 = \left(\int k(x) x^2 dx \right)^2 \int_0^1 (f''(x))^2 / f(x) dx.$$

Moreover, the following law of large numbers for S_n is immediate from corollary 2.7 in Falk (1985) and lemma 2.

Proposition 4. Under the assumptions of theorem 3

$$\alpha_n^{-4} S_n \xrightarrow{n \in \mathbb{N}} c^2/4 \text{ in probability.}$$

Proof of Lemma 2. Write

$$(5) \quad S_n$$

$$= (n-1)^{-2} \sum_{j=1; j \neq 1}^n \sum_{i=1; i \neq j}^{n-1} \tilde{U}_n(X_i, X_j, X_1) + (n-1)^{-2} \sum_{j=1}^n \sum_{i=1; i \neq j}^n \tilde{g}_n(X_i, X_j)$$

where

$$\begin{aligned} & \tilde{U}_n(X_i, X_j, X_1) \\ &= \left\{ \alpha_n^{-1} k((X_i - X_j)/\alpha_n) / f(X_i) - 1 \right\} \left\{ \alpha_n^{-1} k((X_i - X_1)/\alpha_n) / f(X_i) - 1 \right\} \cdot 1_{[0,1]}(X_i) \end{aligned}$$

and

$$\tilde{g}_n(X_i, X_j) = \left\{ \alpha_n^{-1} k((X_i - X_j)/\alpha_n) / f(X_i) - 1 \right\}^2 \cdot 1_{[0,1]}(X_i).$$

Notice that with $j \neq 1, j \neq i \neq 1$

$$E(\tilde{U}_n(X_i, X_j, X_1) | X_j, X_1) = U_n(X_j, X_1)$$

and

$$E(\tilde{g}_n(X_i, X_j) | X_j) = g_n(X_j).$$

Now, we will prove in the following that

$$(6) \quad n \alpha_n^{-4} E \left(\left[(n-1)^{-2} \sum_{j,l=1; j \neq l}^n n^{-1} \sum_{i=1; j \neq i \neq l}^n \left\{ \tilde{U}_n(X_i, X_j, X_l) - U_n(X_j, X_l) \right\} \right]^2 \right) \xrightarrow{n \in \mathbb{N}} 0$$

and

$$(7) \quad n \alpha_n^{-4} E \left(\left[(n-1)^{-2} \sum_{i,j=1; i \neq j}^n \left\{ \tilde{g}_n(X_i, X_j) - g_n(X_j) \right\} \right]^2 \right) \xrightarrow{n \in \mathbb{N}} 0.$$

First we prove (6). To this end, write

$$\begin{aligned} & E \left(\left[\sum_{j,l=1; j \neq l}^n \sum_{i=1; j \neq i \neq l}^n \left\{ \tilde{U}_n(X_i, X_j, X_l) - U_n(X_j, X_l) \right\} \right]^2 \right) \\ &= \sum_{j,l=1; j \neq l}^n \sum_{i=1; j \neq i \neq l}^n \sum_{k,m=1; k \neq m}^n \sum_{r=1; k \neq r \neq m}^n E \left(\left\{ \tilde{U}_n(X_i, X_j, X_l) - U_n(X_j, X_l) \right\} \left\{ \tilde{U}_n(X_r, X_k, X_m) - U_n(X_k, X_m) \right\} \right). \end{aligned}$$

Now, if one of the indices i, r in the above product appears only in one factor, then the conditional expectation of the product with respect to the remaining indices vanishes. Hence, in this case the expectation is zero.

Next, suppose that $i = r$. If none of the remaining indices appears twice, then the expectation is of order $O(\alpha_n^8)$ which after some algebraic manipulation, follows from Fubini's theorem and Taylor expansion of f . If exactly one of the remaining indices appears twice, then the expectation is of order $O(\alpha_n^3)$, whereas in the case that two indices appear twice, it is of order $O(\alpha_n^{-2})$.

Next, suppose that $i \neq r$ and that each of them appears twice (otherwise the expectation is zero). Without loss of generality we may assume that $i = k$ and $r = j$. The remaining possible cases are $l \neq m$ and $l = m$, where $l, m \notin \{i, r\}$. In the first case we obtain for the expectation

the bound $O(\alpha_n^3)$, whereas in the second case we obtain $O(\alpha_n^{-2})$.

Altogether we derive

$$(8) \quad E\left[\sum_{j,l=1; j \neq l}^n \sum_{i=1; j \neq i \neq l}^n \left\{ \tilde{U}_n(X_i, X_j, X_l) - U_n(X_i, X_l) \right\}^2\right] \\ = O(n^5 \alpha_n^8 + n^4 \alpha_n^3 + n^3 \alpha_n^{-2} + n^4 \alpha_n^3 + n^3 \alpha_n^{-2}).$$

Since $\alpha_n \xrightarrow{n \in \mathbb{N}} 0$ and $n \alpha_n^3 \xrightarrow{n \in \mathbb{N}} \infty$, (6) is immediate from (8).

Next we prove (7). To this end, write

$$E\left[\sum_{i,j=1; i \neq j}^n \left\{ \tilde{g}_n(X_i, X_j) - g_n(X_j) \right\}^2\right] \\ = \sum_{i,j=1; i \neq j}^n \sum_{l,m=1; l \neq m}^n E\left\{ \left\{ \tilde{g}_n(X_i, X_j) - g_n(X_j) \right\} \left\{ \tilde{g}_n(X_l, X_m) - g_n(X_m) \right\} \right\}.$$

Suppose at first that one of the indices i, l appears only once in the above product. Then again the expectation is zero by conditioning with respect to the remaining indices.

If $i \neq l$ and $m = i, j = l$, then the expectation is of order $O(\alpha_n^{-3})$. If $i = l$ and $j \neq m$, the expectation is of order $O(\alpha_n^{-2})$, whereas in the case $i = l$ and $j = m$ it is of order $O(\alpha_n^{-4})$ (recall that $j \neq i = l \neq m$).

Altogether we obtain

$$(9) \quad E\left[\sum_{i,j=1; i \neq j}^n \left\{ \tilde{g}_n(X_i, X_j) - g_n(X_j) \right\}^2\right] \\ = O(n^2 \alpha_n^{-3} + n^3 \alpha_n^{-2} + n^2 \alpha_n^{-4}).$$

Since $n\alpha_n^3 \rightarrow 0$, (9) implies (7).

The assertion of lemma 2 is now immediate from (6) and (7), the fact that $E(|U_n(X_1, X_2)|) = O(1)$, $E(g_n(X_1)) = O(\alpha_n^{-1})$ and elementary computations.

Remarks. In theorem 2.9 in Falk (1985) asymptotic normality of T_n was established also in the case, where f has the support $[0,1]$ on which it is continuous and bounded. However, the appropriate norming constants in this case differ completely from those which are needed in the case where f is smooth on the real line. This is caused by the fact that a kernel density estimator fails at the endpoints of the underlying density.

It would be interesting to know whether also in this case S_n is asymptotically equivalent to T_n . However, it is easy to see that the proof of lemma 2 fails in this case and thus, this question is still open.

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